SafePath – Autonomous Porter Guide

The robot maps a room with its two ultrasonic sensors, returns home, and then shuttles between a START and an END point like a warehouse porter – avoiding people or objects that appear on the way. **No new hardware and no wiring changes** are needed.

How the system is split

Robot (ESP32) – muscles & reflexes	Laptop dashboard – the brain
• Creates the Wi-Fi network • Runs short commands: drive, turn, scan • **Emergency brake** if anything is closer than 25 cm – the servo sweeps left/centre/right while driving, like a wide bumper • **Stall detection**: if wheels turn but front AND back distances don't change, it stops • Gentle soft-start; stops by itself if the laptop goes silent	• Builds the map from **both** sensors (every scan and every move) • Reverses with the back sensor instead of spinning 180° • Backs away from walls before turning • Explores unknown areas automatically • Plans the shortest safe path (A* search) • Picks an END point automatically • Waits / detours when an obstacle appears • Corrects drift using the walls

Files

File	What to do
firmware/safepath_auto/safepath_auto.ino	Upload to the ESP32 (library: ESP32Servo). Replaces the old code; the phone page still works.
dashboard/index.html (+ style.css, js/)	Open on the laptop. Keep the folder together. Works without internet.

Step by step

#	Step	Details
1	Upload	Upload safepath_auto.ino over USB. Unplug USB, switch the battery on.
2	Connect	Laptop joins Wi-Fi SafePath_Robot (SafePath123). Open index.html in Chrome/Edge, click Connect. If asked to allow local network access, click Allow.
3	Calibrate	Calibration & settings → 'Test: drive 2 s', measure the distance in cm, Apply. 'Test: turn 90°', enter the real angle, Apply. Repeat once to fine-tune.
4	Home mark	Put tape on the floor. Place the robot on it facing the direction you call 'ahead'.
5	Map	Click Initiate mapping. The robot scans, explores and comes back home (a few minutes).
6	Start / End	Start = home. End is chosen automatically (farthest safe spot). Change with Set start / Set end.
7	Shuttle	Set Trips (0 = forever) and Wait time, click Start shuttle.
8	Save	Map file → Save map. Next time: Load map, place robot on the home mark, Robot is at home.

What happens when something blocks the path

1. The robot brakes by itself (on the ESP32 – no Wi-Fi delay).
2. The laptop waits about 4 seconds – a person may just walk past.
3. If it is still there, that spot is marked **temporary** (orange) and a detour is planned.
4. Temporary obstacles are forgotten after 30 s, so the normal route is used again once the way is clear.

Accuracy without wheel sensors

BO motors have no encoders, so distance and turns are worked out from **time**, which drifts. The dashboard fights this in three ways: it squares itself to walls by comparing left/right echoes, it corrects its position using known wall distances, and it re-checks its position at every START/END stop. It also keeps a safety gap from

walls when planning.

For a good demo: use a small area (about 3×3 m) with straight walls (cardboard works), a smooth floor, a charged battery, and calibrate first. If the robot ever looks confused: press STOP, place it on the home mark facing ahead, click **Robot is at home**, and continue.

Sensor mounting (important)

• **Back sensor (6.8 cm high)** is low: the wide beam can bounce off the floor and give false short readings. Raise it to about 10 cm like the front one, or tilt it up by 5–10°.
• Both sensors should point level and straight, not down.
• Ultrasonic sensors struggle with **thin chair legs**, soft fabric and objects at an angle. The map now marks echoes across the beam width and the robot sweeps while driving, but for demos prefer boxes / solid obstacles.

Troubleshooting

Problem	Fix
Connect fails	Laptop on SafePath_Robot Wi-Fi? Firmware uploaded? Try Edge/Firefox, or run 'python -m http.server 8000' in the dashboard folder and open http://localhost:8000
Turns too far / not enough	Redo the turn calibration. Check both wheels grip the floor.
Robot curves when driving straight	Lower LEFT_TRIM or RIGHT_TRIM (faster side) in the firmware, e.g. 92.
Map looks smeared	Calibrate again, slow the robot (Drive power 150–170), use a smaller area.
Keeps disconnecting while running	Usually a power dip resetting the ESP32 when motors + servo start. Charge the battery, put the 470uF capacitor across 5 V/GND near the ESP32, keep motor wires away from the ESP32. The dashboard now retries and says 'robot restarted' if this happens. Keep the laptop within ~3 m.
Still too fast	Lower Drive power / Turn power (try 125–135), then redo both calibration tests.
Hits things while spinning	Raise Emergency stop to 30 cm. The robot backs off first if a wall is under 22 cm.